

Using SSM and Amazon Inspector

(LAB-M04-01)

Overview

Patch management is an ongoing task that requires many resources as your server fleet grows if management is not automated. AWS Systems Manager is a scalable tool for remote management of your fleet of Amazon Elastic Compute Cloud (Amazon EC2) instances.

In this lab, you will use AWS Systems Manager to perform automated patch delivery. You will use Systems Manager Run Command to install the Amazon Inspector agent on a fleet of instances. The agent enables EC2 instances to communicate with Amazon Inspector, which will run security assessments on your EC2 instances. Once the assessments are complete, Amazon Inspector will create and deploy a baseline patch using the Systems Manager Automation feature to bring your instances into compliance.

Objectives

After completing this lab, you will be able to:

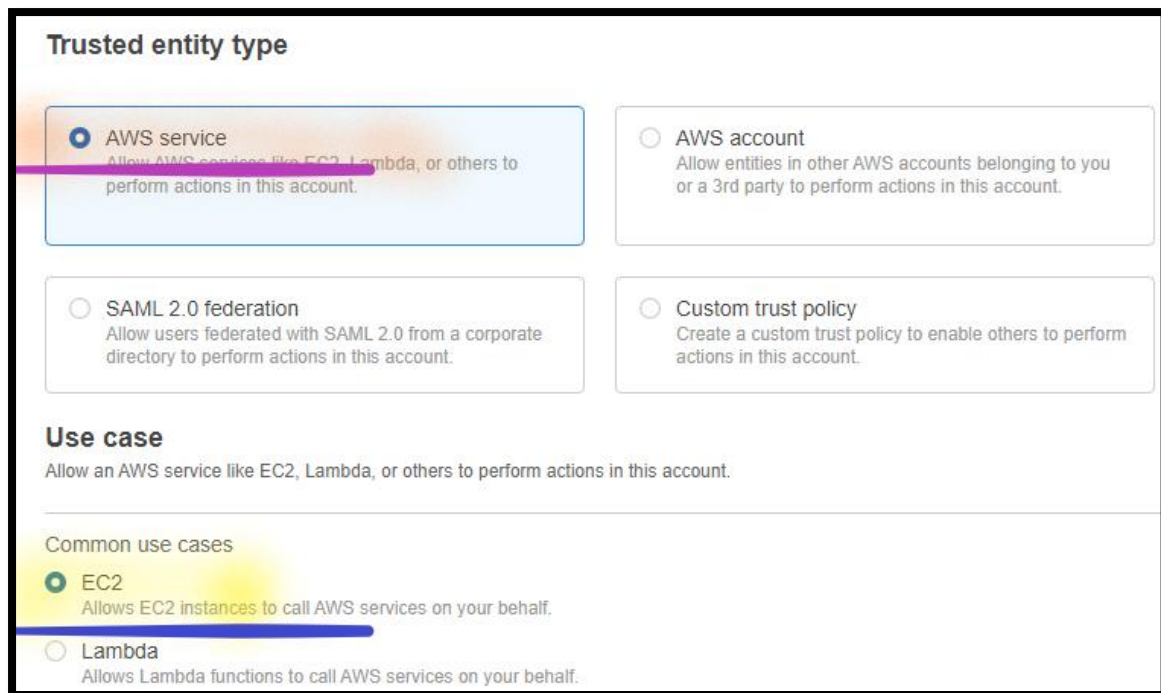
- Run commands on a group of Amazon EC2 instances using Systems Manager Run Command
- Set up and run an Amazon Inspector scan
- Deploy patches to instances via Systems Manager Automation

Task 1: Create IAM Role

In this task, you will create IAM Role for EC2 instances.

Step 1: Create IAM Roles for AWS System Manager

1. In the **AWS Management Console**, on the **Services** menu, click **IAM**.
2. Select **Roles**, click on **Create role**.
 - a. In the **Select trusted entity** section:
 - i. **Trusted entity type**: Select **AWS service**.
 - ii. **Common use cases**: Select **EC2**.



Trusted entity type

☒ **AWS service**
Allow AWS services like EC2, Lambda, or others to perform actions in this account.

☐ **AWS account**
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.

☐ **SAML 2.0 federation**
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.

☐ **Custom trust policy**
Create a custom trust policy to enable others to perform actions in this account.

Use case
Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Common use cases

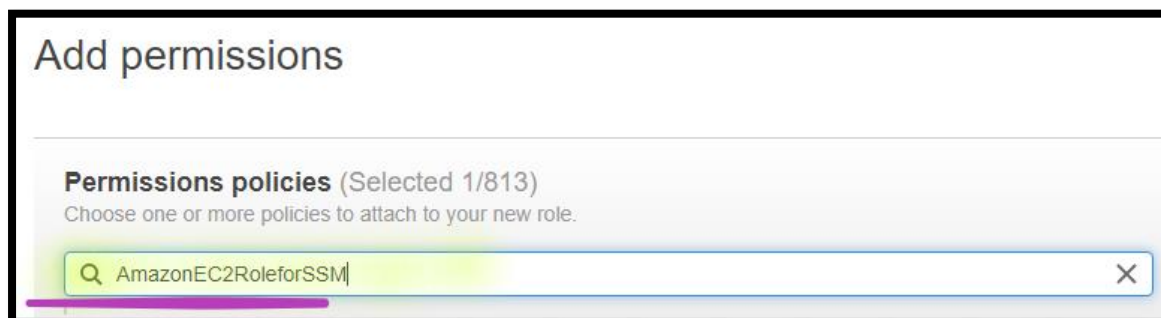
☒ **EC2**
Allows EC2 instances to call AWS services on your behalf.

☐ **Lambda**
Allows Lambda functions to call AWS services on your behalf.

iii. Select **Next**.

b. In the **Add Permissions** section:

i. in the **Search box**, type **AmazonEC2RoleforSSM**, press **Enter**.

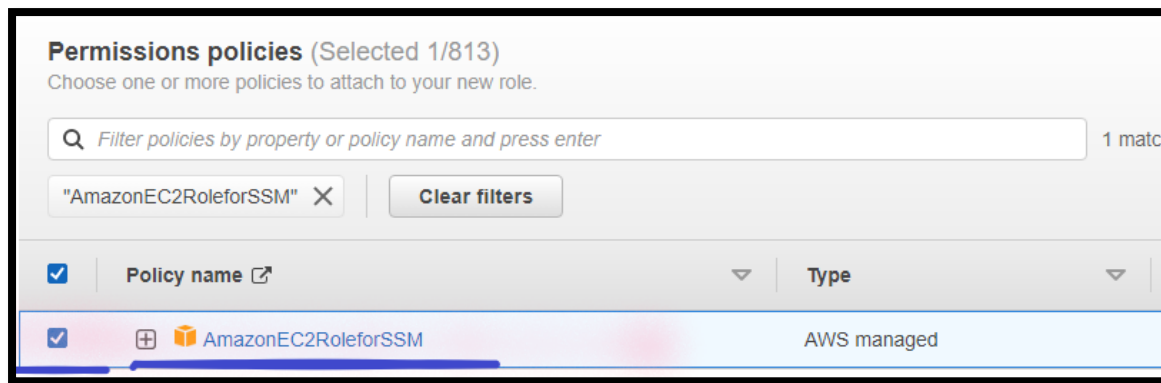


Add permissions

Permissions policies (Selected 1/813)
Choose one or more policies to attach to your new role.

Q AmazonEC2RoleforSSM X

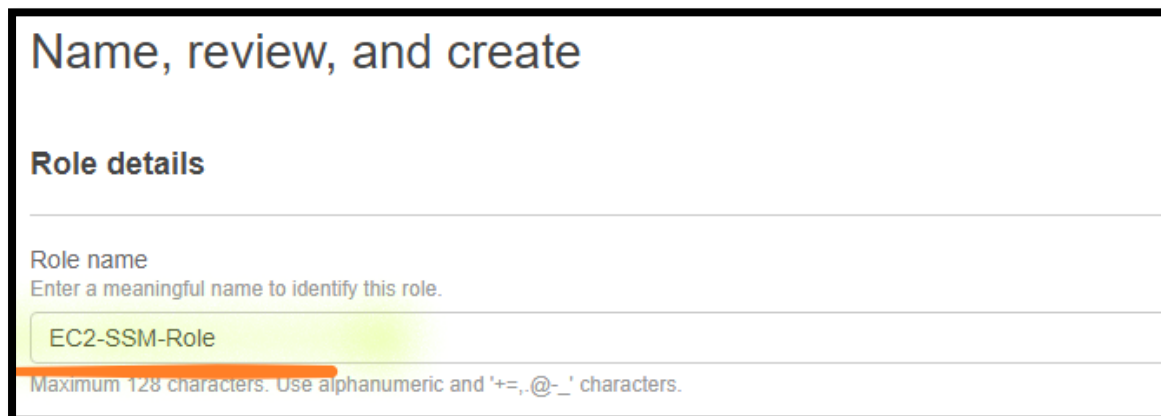
ii. Select **AmazonEC2RoleforSSM**.



iii. Select **Next**.

c. In the **Name, review, and create** section:

i. **Role Name:** Write **EC2-SSM-Role**.



ii. Click **Create role**.

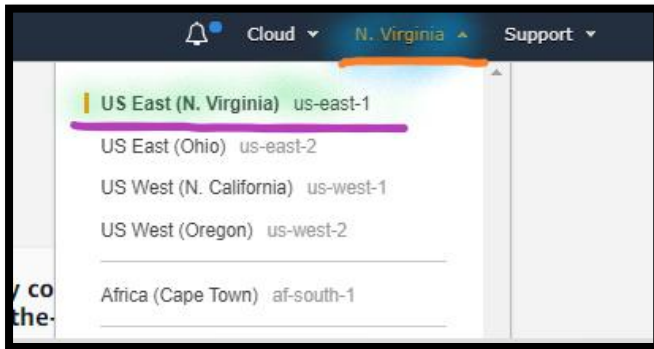
Note: **Wait**, till Role **EC2-SSM-Role** gets created.

Task 2: Deploy Virtual Machines

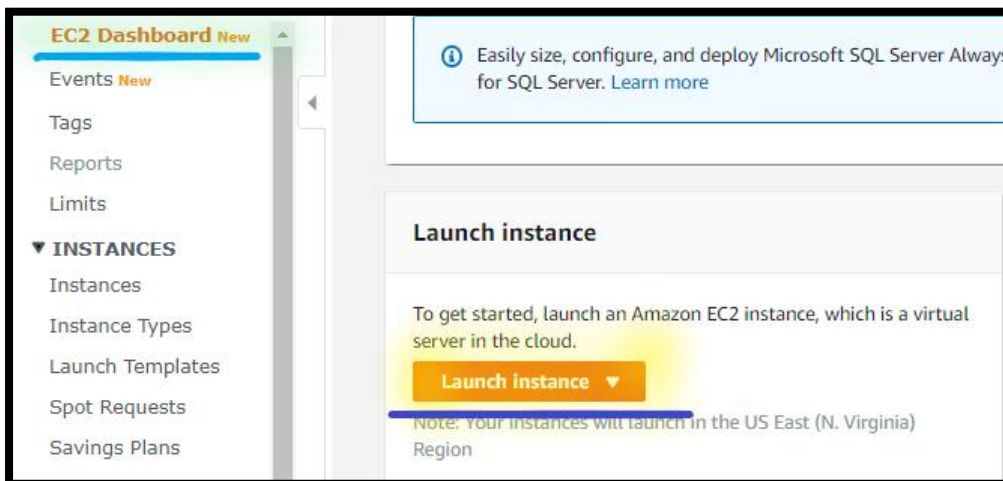
In this task, you will create two EC2 instances, Windows and Linux.

Step 1: Create Windows Instance

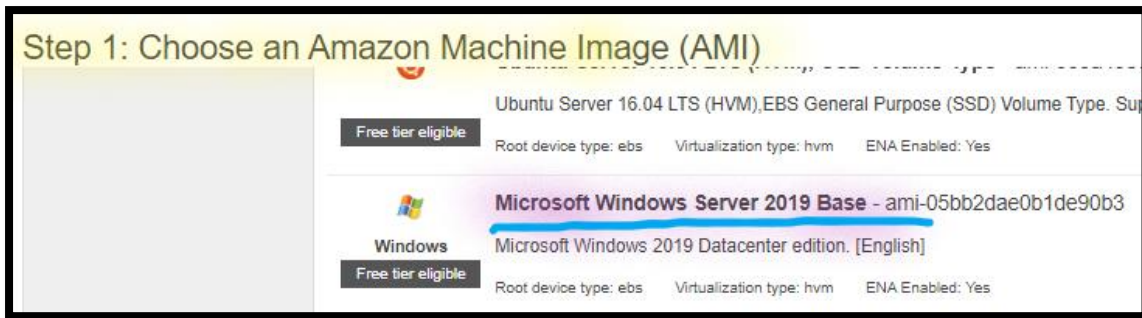
1. Dropdown and select **US East (N. Virginia)** region.



2. In the **AWS Management Console**, on the **Services** menu, click **EC2**.
3. Click **Launch Instance**.



- a. In the **Choose an Amazon Machine Image (AMI)** page:
 - i. Select the **Microsoft Windows Server 2019 Base**.
 - ii. Click **Select**.

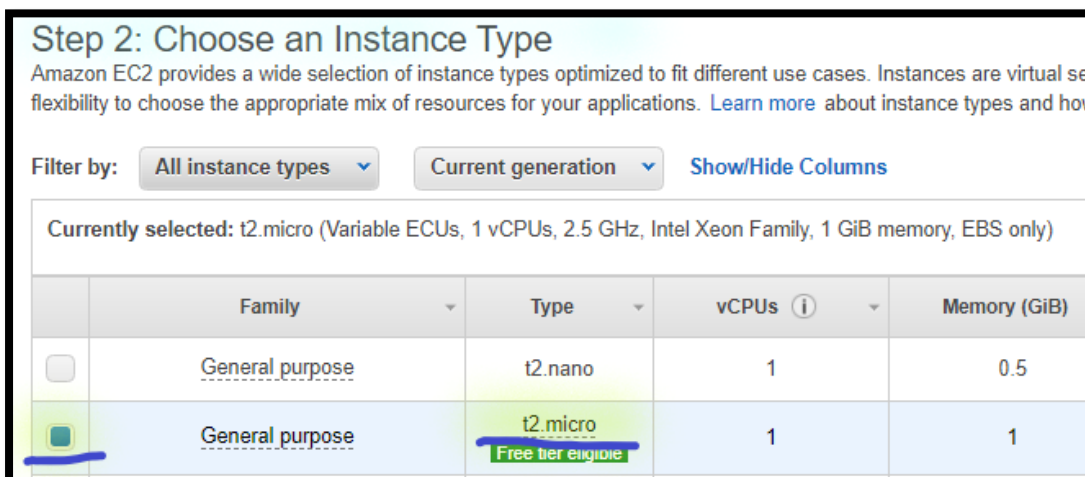


b. In the **Choose an Instance Type** page:

i. Select the **t2.micro**.

Note: This instance type is covered within the free tier.

ii. Click **Next: Configure Instance Details**.



c. In the **Configure Instance Details** page:

i. **IAM Role:** Dropdown and select **EC2-SSM-Role**.



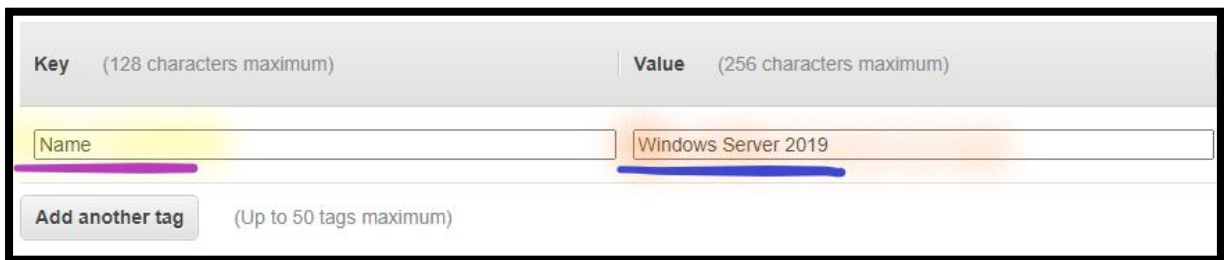
Note: Leave the other settings as default.

ii. Select **Next: Add Storage**.

d. In the **Add Storage** page:

Note: Leave all the settings as default.

- i. Select **Next: Add Tags**.
- e. In the **Add Tags** page:
 - i. Click on **Add Tag**:
 - a) **Key**: Write **Name**.
 - b) **Value**: Write **Windows Server 2019**.

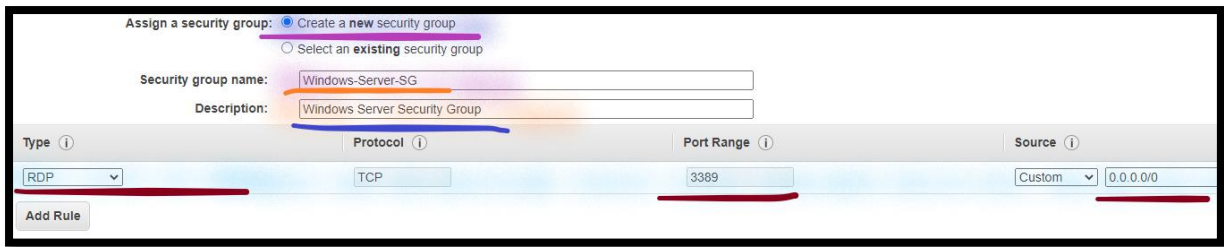


The screenshot shows a web interface for adding tags. It has two input fields: 'Key' (128 characters maximum) and 'Value' (256 characters maximum). The 'Key' field contains the text 'Name' and the 'Value' field contains 'Windows Server 2019'. Below these fields is a button labeled 'Add another tag' and a note '(Up to 50 tags maximum)'.

- ii. Select **Next: Configure Security Group**.
- f. In the **Configure Security Group** page:
 - i. **Assign a security group**: Select **Create a new security group**.
 - a) **Security group name**: Write **Windows-Server-SG**.
 - b) **Description**: Write **Windows Server Security Group**.

Info: RDP Port [3389], already added and allowed from anywhere [0.0.0.0/0].

Note: Leave the other settings as default.



- ii. Select **Review and Launch**.

Info: On the Review Instance Launch page, check the details of your instance, and make any necessary changes by choosing the appropriate Edit link.

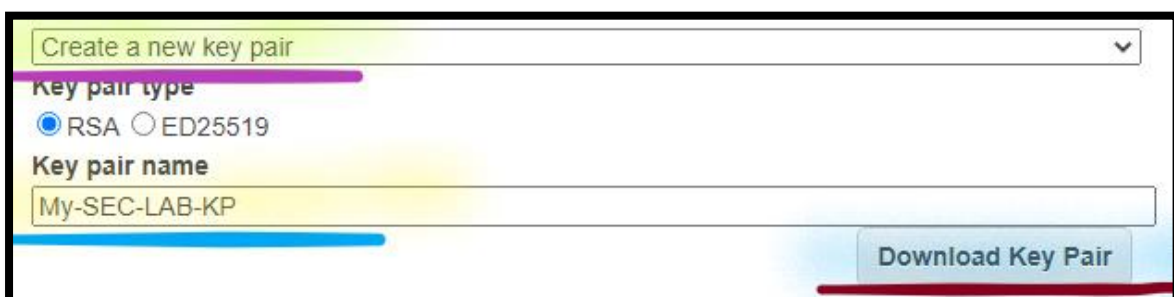
- g. In the **Review Instance Launch** page:

- i. Select **Launch**.

- a) In the **Select an existing key pair or create a new key pair** section:

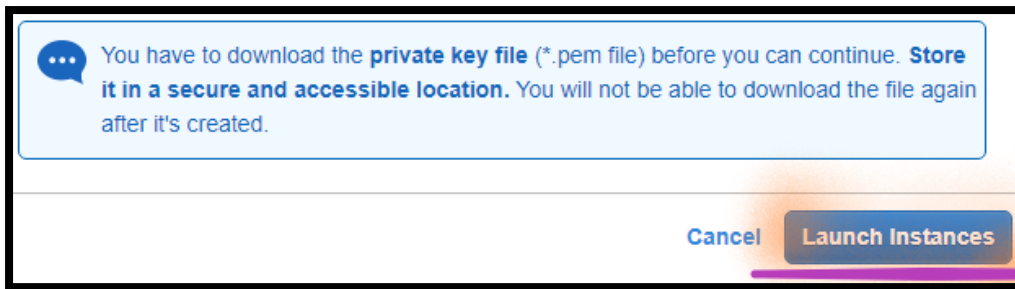
- 1) **First selection box:** Dropdown and select **Create a new key pair**.
- 2) **Key pair name:** Write **My-SEC-LAB-KP**.
- 3) Click on **Download Key Pair**.

Note: **My-SEC-LAB-KP.pem** will be downloaded to your computer and make sure to save this key pair in a safe location on your computer.



Note: Leave the other settings as default.

- i. Click on **Launch Instances**.



- h. Select the **View Instances**.

Step 2: Create Ubuntu Instance

4. Click **Launch Instance**.
 - a. In the **Choose an Amazon Machine Image (AMI)** page:
 - i. Select the **Ubuntu Server 18.04 LTS**.
 - ii. Click **Select**.
 - b. In the **Choose an Instance Type** page:
 - i. Select the **t2.micro**.
 - ii. Click **Next: Configure Instance Details**.
 - c. In the **Configure Instance Details** page:
 - i. **IAM Role**: Dropdown and select **EC2-SSM-Role**.

Note: Leave the other settings as default.

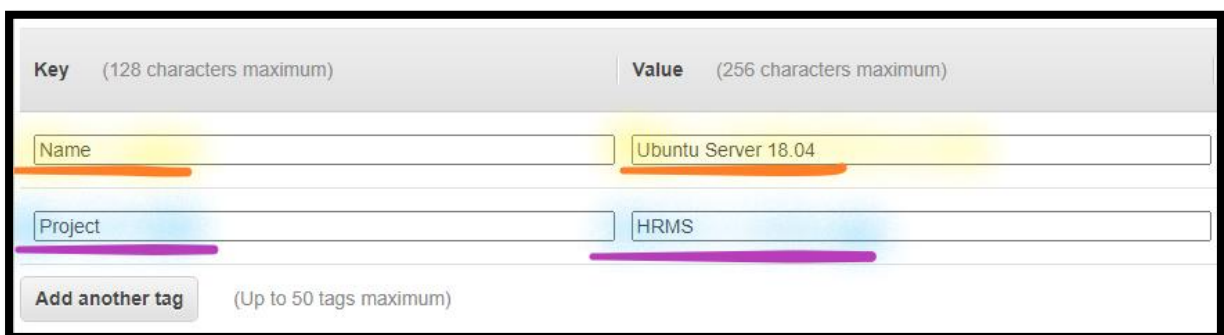
- ii. Select **Next: Add Storage**.
 - d. In the **Add Storage** page:

Note: Leave all the settings as default.

- i. Select **Next: Add Tags**.

e. In the **Add Tags** page:

- i. Click on **Add Tag**:
 - a) **Key**: Write **Name**.
 - b) **Value**: Write **Ubuntu Server 18.04**.
- ii. Click on **Add Tag**:
 - a) **Key**: Write **Project**.
 - b) **Value**: Write **HRMS**.



The screenshot shows a form titled 'Add Tags' with two columns: 'Key' (128 characters maximum) and 'Value' (256 characters maximum). The first row has 'Name' in the Key field and 'Ubuntu Server 18.04' in the Value field. The second row has 'Project' in the Key field and 'HRMS' in the Value field. Below the form is a button labeled 'Add another tag' and a note '(Up to 50 tags maximum)'.

iii. Select **Next: Configure Security Group**.

f. In the **Configure Security Group** page:

- i. **Assign a security group**: Select **Create a new security group**.
 - a) **Security group name**: Write **Ubuntu-Server-SG**.
 - b) **Description**: Write **Ubuntu Server Security Group**.

Info: SSH Port [22], already added and allowed from anywhere [0.0.0.0/0].

Note: Leave the other settings as default.

ii. Select **Review and Launch**.

g. In the **Review Instance Launch** page:

- i. Select **Launch**.

- a) In the **Select an existing key pair or create a new key pair** section:
- 1) **First selection box:** Dropdown and select **Choose an existing key pair**.
 - 2) **Key pair name:** Dropdown and select **My-SEC-LAB-KP**.
 - 3) **Enable** the **Checkmark** against the **I acknowledge that I have access ...**

Note: Leave the other settings as default.

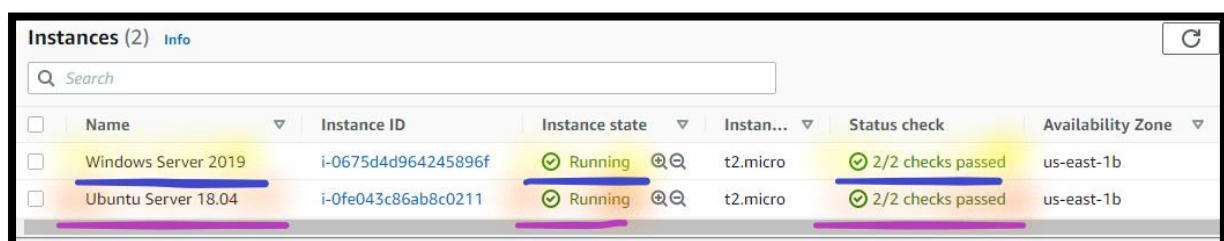
- ii. Click on **Launch Instances**.

- h. Select the **View Instances**.

Step 3: Check the Windows and Ubuntu Server Status

5. In the **AWS Management Console**, on the **Services** menu, click **EC2**.
6. Click **Instance**.
7. Select **Windows Server 2019** and **Ubuntu Server 18.04**.
 - a. **Wait** for the **Instance State** to change to **Running** state.
 - b. **Wait** for the **Status check** to change to **2/2 checks passed**.

Note: **Wait** (~5 mnts.) for the server deployment.



	Name	Instance ID	Instance state	Instan...	Status check	Availability Zone
☐	Windows Server 2019	i-0675d4d964245896f	Running	t2.micro	2/2 checks passed	us-east-1b
☐	Ubuntu Server 18.04	i-0fe043c86ab8c0211	Running	t2.micro	2/2 checks passed	us-east-1b

Task 3: Create AWS S3 Bucket

Step 1: Create a Bucket

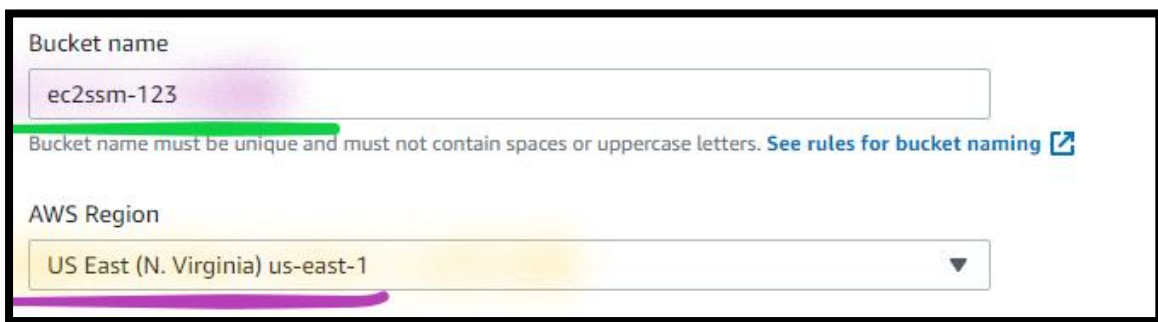
8. In the **AWS Management Console**, on the **Services** menu, click **S3**.

9. Click **Create bucket**.

a. **Bucket name**: Write **ec2ssm-123**.

Note: Replace **123** with a random number to make bucket name unique.

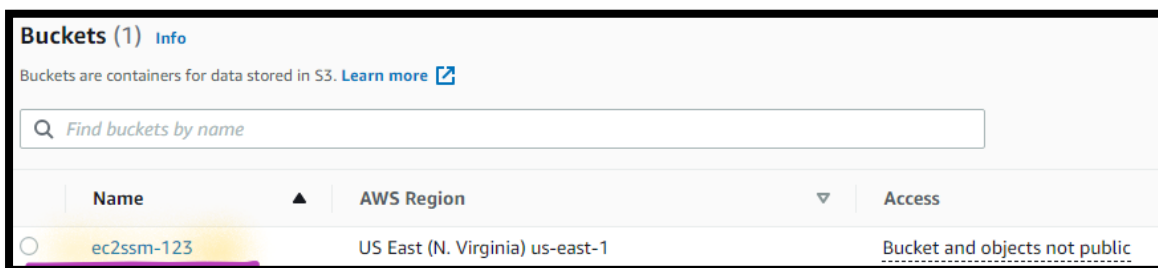
b. **Region**: Dropdown and Select **US East (N. Virginia)**.



Note: Leave the other settings as default.

c. Click **Create bucket**.

Note: You can see the Bucket details.



Name	AWS Region	Access
ec2ssm-123	US East (N. Virginia) us-east-1	Bucket and objects not public

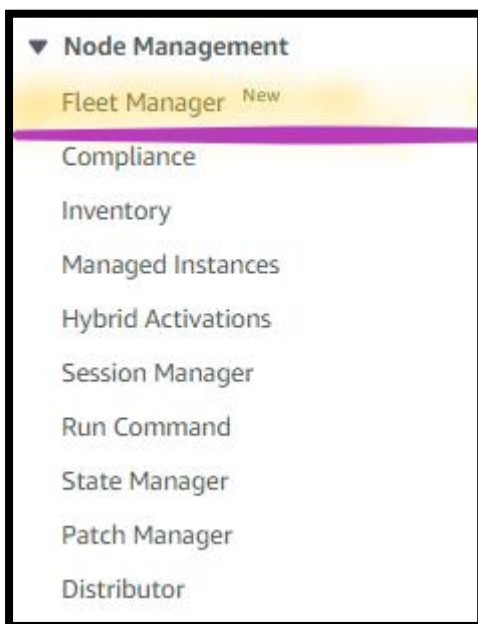
Task 4: Scan and Install the Patches On-Demand

In this task using the Patch manager, you can scan using the Patch Manager, provided predefined patch baselines and run on-demand patching operations without creating a schedule in order to update the compliance status of your managed nodes or to install patches on noncompliant nodes.

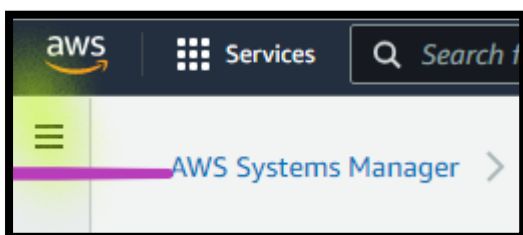
Patch Manager has a pre-defined patch baseline that approves all patches classified as critical updates or security updates with a severity of Critical or Important. These patches are automatically approved by this baseline seven days after they are released by Microsoft.

Step 1: View the Host Details

10. Dropdown and select **US East (N. Virginia)** region.
11. In the **AWS Management Console**, on the **Services** menu, click **Systems Manager**.
 - a. Select **Fleet Manager** under **Node management**.

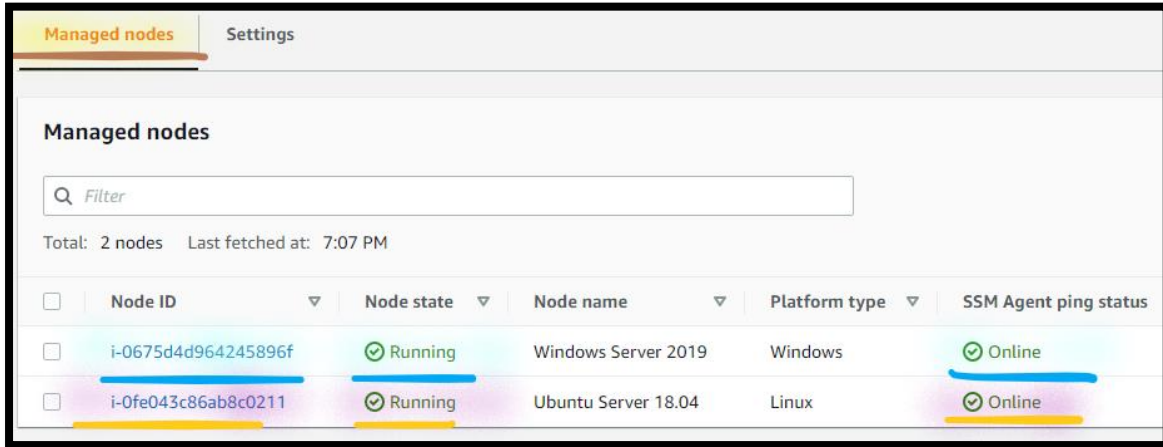


Note: If you are not seeing Fleet manager option, click on **three lines** on the left side.



- i. Click on **Managed Instances**.

Note: You can see the **Window server**.

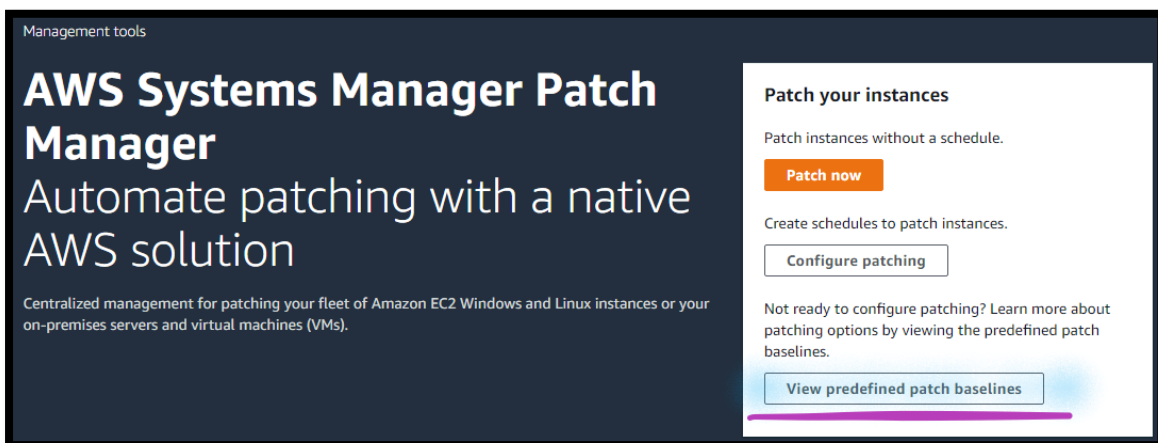


	Node ID	Node state	Node name	Platform type	SSM Agent ping status
☐	i-0675d4d964245896f	Running	Windows Server 2019	Windows	Online
☐	i-0fe043c86ab8c0211	Running	Ubuntu Server 18.04	Linux	Online

Step 2: View the Baseline Patches

12. From the **Systems Manager** Page:

- a. Select **Patch Manager** under **Node management**.
- b. Select **View predefined patch baselines**.



Management tools

AWS Systems Manager Patch Manager

Automate patching with a native AWS solution

Centralized management for patching your fleet of Amazon EC2 Windows and Linux instances or your on-premises servers and virtual machines (VMs).

Patch your instances

Patch instances without a schedule.

Patch now

Create schedules to patch instances.

Configure patching

Not ready to configure patching? Learn more about patching options by viewing the predefined patch baselines.

View predefined patch baselines

Note: You can see all the **Baseline patches**.

Patch baselines					
View details Edit Delete Actions Create patch baseline					
Baseline ID	Baseline name	Description	Operating system	Default baseline	
☐ pb-03e3f588ec25344c	AWS-CentOSDefaultPatchBaseline	Default Patch Baseline for CentOS Provided by AWS.	CentOS	Yes	
☐ pb-06bfff38e95fe85c02	AWS-OracleLinuxDefaultPatchBaseline	Default Patch Baseline for Oracle Linux Server Provided by AWS.	Oracle Linux	Yes	
☐ pb-07d8884178197b66b	AWS-SuseDefaultPatchBaseline	Default Patch Baseline for Suse Provided by AWS.	SUSE	Yes	
☐ pb-096d816473f2bdb03	AWS-WindowsPredefinedPatchBaseline-OS	Approves all Windows Server operating system patches that are classified as CriticalUpdates or SecurityUpdates and that have an MSRC severity of Critical or Important. Patches are auto-approved seven days after release.	Windows	No	
☐ pb-09a5f8eb62bde80b1	AWS-DebianDefaultPatchBaseline	Default Patch Baseline for Debian Provided by AWS.	Debian	Yes	
☐ pb-09ca3fb51f0412ec3	AWS-DefaultPatchBaseline	Default Patch Baseline Provided by AWS.	Windows	Yes	

i. Open the **Windows baseline** Patches.

Note: You can see the **Baseline patches** details.

Baseline ID: pb-09ca3fb51f0412ec3

Description

Baseline ID

arn:aws:ssm:us-east-1:075727635805:patchbaseline/pb-09ca3fb51f0412ec3

Description

Default Patch Baseline Provided by AWS.

Default baseline

Yes

Created date (UTC)

Tue, 01 May 2018 17:13:20 GMT

Baseline name

AWS-DefaultPatchBaseline

Operating system

Windows Server

Patch groups

-

Modified date (UTC)

Tue, 01 May 2018 17:13:20 GMT

Approval rules

Product	Classification	Severity	Auto approval delay
-	CriticalUpdates,SecurityUpdates	Critical,Important	Wait 7 days before approving

Info: **Classification** On the Review Instance Launch page, check the details of your instance, and make any necessary changes by choosing the appropriate Edit link.

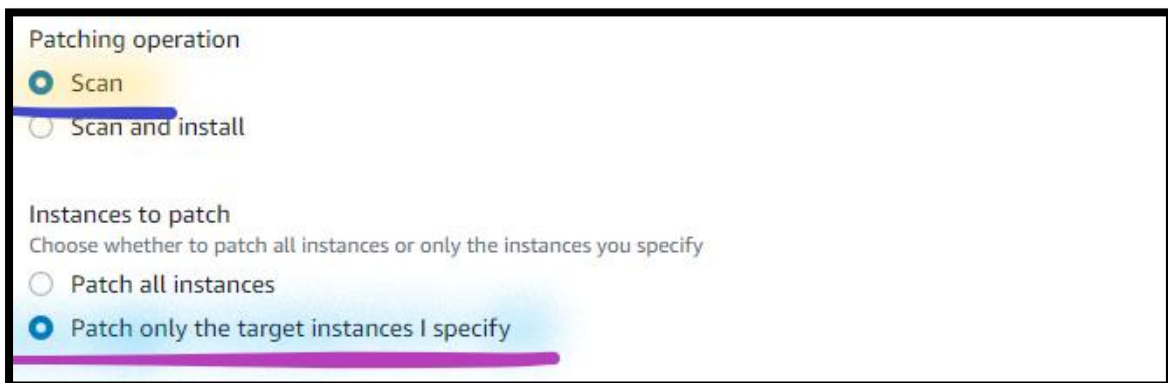
Info: **Auto approval delay** On the Review Instance Launch page, check the details of your instance, and make any necessary changes by choosing the appropriate Edit link.

- ii. Open and view the **Ubuntu baseline** Patches.

Step 3: Scan the Patches On-Demand

13. From the **Systems Manager** Page:

- a. Select **Patch Manager** under **Node management**.
- b. Select **Patch now**.
- c. From the **Patch instances now** page:
 - i. **Patching operation:** Select **Scan**.
 - ii. **Instances to patch:** Select **Patch only the target instances I specify**.



Patching operation

☒ Scan

☐ Scan and install

Instances to patch

Choose whether to patch all instances or only the instances you specify

☐ Patch all instances

☒ Patch only the target instances I specify

- iii. **Target selection:** Select **Choose instances manually**.
 - a) **Instances:** Select **Windows Server 2019** and **Ubuntu Server 18.04** servers.

Target selection
Choose a method for selecting targets.

☐ Specify instance tags
Specify one or more tag key-value pairs to select instances that share those tags.

☒ **Choose instances manually**
Manually select the instances you want to register as targets.

☐ Choose a resource group
Choose a resource group that includes the target.

Instances

☒	Node ID	Source type	Source ID	Name	Ping status	Node state
☒	i-0675d4d964245896f	AWS::EC2::Instance	i-0675d4d964245896f	Windows Server 2019	Online	running
☒	i-0fe043c86ab8c0211	AWS::EC2::Instance	i-0fe043c86ab8c0211	Ubuntu Server 18.04	Online	running

- iv. **Patching log storage:** Dropdown and select **ec2ssm-123** S3 bucket.

Patching log storage New

Select or create an S3 bucket for storing patching operation logs. Select **Do not store logs** if you don't require log information.

- v. Select **Patch now**.

Note: You can see the **Execution details are loading**.

Note: **Wait**, till **Operation gets Successful**.

Operation was Successful

AWS Systems Manager > Patch Manager > Association execution summary

Association execution summary

AWS-PatchNowAssociation

Association ID 54cc0251-fc51-4455-ac51-d1315a98ea8a	Execution ID 4377e571-a617-4afb-81e7-9c29163bfed6
Status Success	Operation Scan
Reboot option NoReboot	Targets InstanceIds: i-0675d4d964245896f, i-0fe043c86ab8c0211

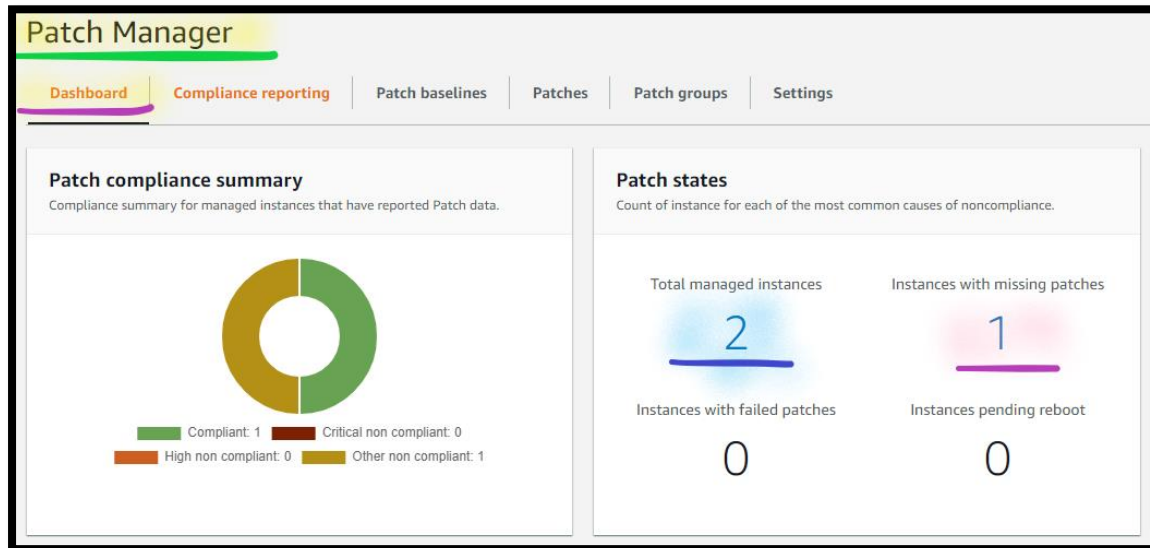
Step 4: View the Patches Compliance

14. From the **Systems Manager** Page:

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- Select **Patch Manager** under **Node management**.
- Select **Dashboard**.

Note: You can see the **Patch states** details.



- Select **Compliance Reporting**.

Note: You can see the **Compliance status** details.

Note: You can also see the **Patch baseline** used for Scanning against Instance **OS type**.

The screenshot shows the Compliance Reporting page with the following table:

	Name	Instance ID	Compliance sta...	Security noncompliant count	Baseline ID used
☐	Windows Server 2019	i-0675d4d964245896f 🔗	✅ Compliant	0	pb-09ca3fb51f0412ec3 🔗
☐	Ubuntu Server 18.04	i-0fe043c86ab8c0211 🔗	❌ Not compliant	34 🔗	pb-0c7e89f711c3095f4 🔗

- Click on the **Non-Compliant count** against **Ubuntu Server**:

Note: You can see the **Missing patches** details.

Patch summary (34)

Search for Patches

State: Not equal: installed X State: Not equal: installedother X Classification: Begin with: Security X Clear filters

	Name	State	Classification	Severity	Compliance level	Installed time
☐	busybox-initramfs.amd64	Missing	Security	important	UNSPECIFIED	-
☐	busybox-static.amd64	Missing	Security	standard	UNSPECIFIED	-
☐	libc-bin.amd64	Missing	Security	required	UNSPECIFIED	-
☐	libc6.amd64	Missing	Security	required	UNSPECIFIED	-

Step 5: View the Compliance Dashboard

15. From the **Systems Manager Page**:

- a. Select **Compliance** under **Node management**.
 - i. **Group dashboard results based on:** Select **Compliance type**.
 - ii. Click on **Non-Compliant resources**.

Group dashboard results based on

☒ Compliance type ☐ Patch group ☐ Res

Filter further

Compliance resources summary

Compliance type	Compliant resources	Non-Compliant resources	Critical resources	High resources
Association	✓ 2	⚠ 0	⚠ 0	⚠ 0
Patch	✓ 1	⚠ 1	⚠ 0	⚠ 0

Note: You can see the **Non Compliant** resources details.

- e. Select the **Resource ID**.

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Note: You can see the **Non-compliance** details.

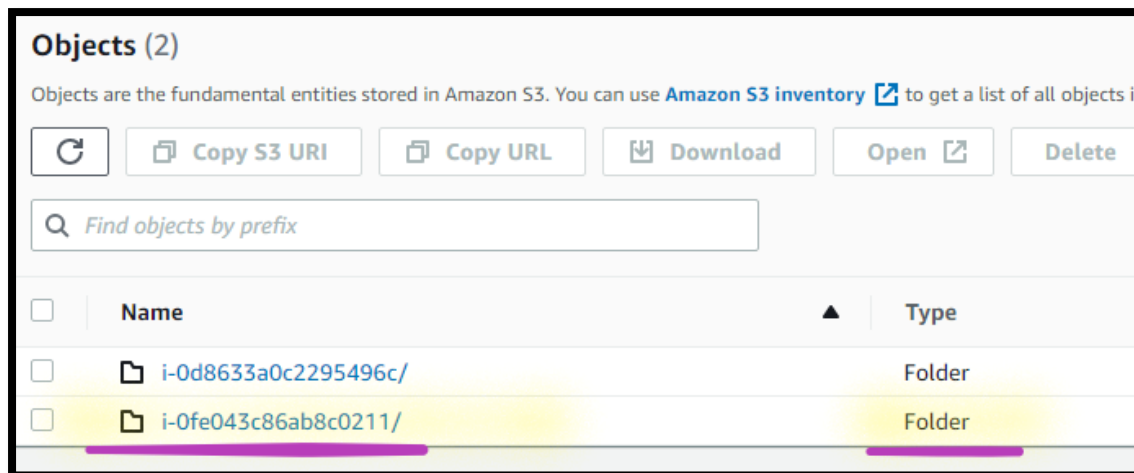
Resource					
Status : Equal : Non-compliant		ComplianceType : Equal : Patch		Clear Filters	
ID	Resource type	Compliance type	Overall severity	Overall status	Execution
i-0fe043c86ab8c0211	ManagedInstance	Patch	Unspecified	Non-compliant	Sat, 05 M
Compliance rule					
Q			All	Non-compliant	
Status : Equal : Non-compliant		ComplianceType : Equal : Patch		Severity : Equal : All	ResourceId : Equal : i-0fe043c86ab8c0211
				Clear Filters	
ID	Compliance type	Resource ID	Severity	Status	Execution time
busybox-initramfs.amd64	Patch	i-0fe043c86ab8c0211	Unspecified	Non-compliant	Sat, 05 Mar 2022 14:45:36 GMT
busybox-static.amd64	Patch	i-0fe043c86ab8c0211	Unspecified	Non-compliant	Sat, 05 Mar 2022 14:45:36 GMT
libc-bin.amd64	Patch	i-0fe043c86ab8c0211	Unspecified	Non-compliant	Sat, 05 Mar 2022 14:45:36 GMT
libc6.amd64	Patch	i-0fe043c86ab8c0211	Unspecified	Non-compliant	Sat, 05 Mar 2022 14:45:36 GMT
libexpat1.amd64	Patch	i-0fe043c86ab8c0211	Unspecified	Non-compliant	Sat, 05 Mar 2022 14:45:36 GMT

Step 6: View the Scan Log Details

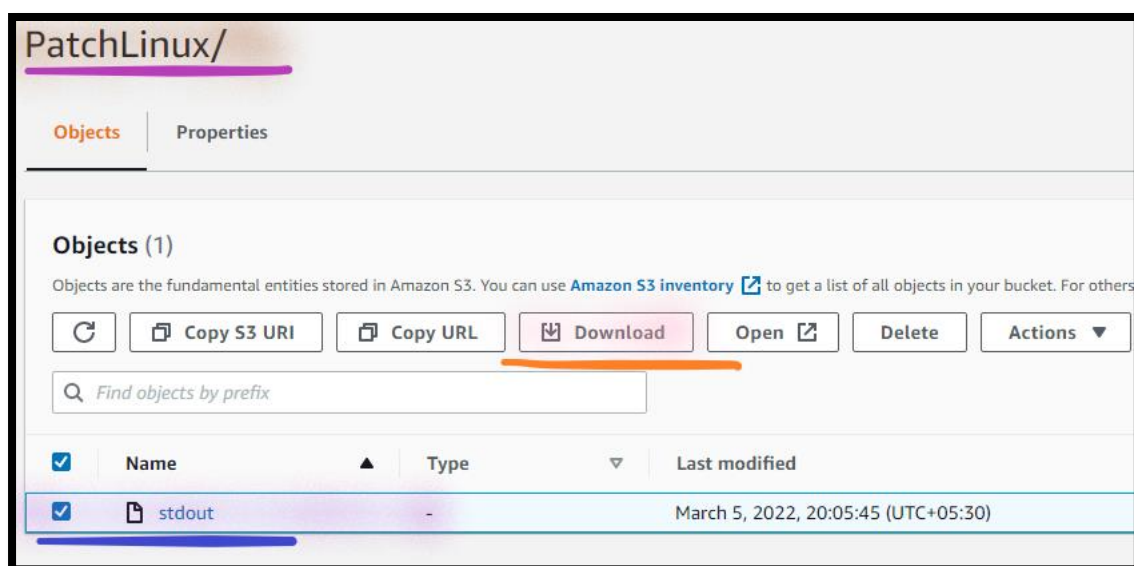
16. In the **AWS Management Console**, on the **Services** menu, click **S3**.

17. Select **Buckets**.

- Open the **ec2ssm-123** bucket.
- Open the **AWS-PatchNow/** folder.
- Open the **Sub folder**.
- Open the **Sub folder** for one of the **Instance Id**.



- e. Open the **PatchLinux/** folder.
- f. Select the **stdout** object.
- g. Select the **Download**.



Note: **stdout** file gets downloaded in the **local Desktop/ Laptop**.

Note: You can **Open** the **stdout** file in the **Notepad** to view the details.

```

stdout.txt - Notepad
File Edit Format View Help
/usr/bin/python3
/usr/bin/apt-get
Reading package lists...
Building dependency tree...
Reading state information...
python3-apt is already the newest version (1.6.5ubuntu0.7).
0 upgraded, 0 newly installed, 0 to remove and 48 not upgraded.
Using python binary: 'python3'
Using Python Version: Python 3.6.9
/usr/bin/curl
/usr/bin/wget
03/05/2022 14:35:11 root [INFO]: Downloading payload from https://s3.dualstack.us-east-1.amazonaws.com/av
03/05/2022 14:35:12 root [INFO]: Attempting to import entrance file os_selector
03/05/2022 14:35:12 root [INFO]: Running with snapshot id =  and operation = Scan
03/05/2022 14:35:12 root [INFO]: Found credentials in environment variables.
03/05/2022 14:35:13 root [INFO]: Instance Id: i-0fe043c86ab8c0211
03/05/2022 14:35:13 root [INFO]: Region: us-east-1
03/05/2022 14:35:13 root [INFO]: Product: Ubuntu18.04
03/05/2022 14:35:13 root [INFO]: Patch Group:
03/05/2022 14:35:13 root [INFO]: Operation type: Scan
03/05/2022 14:35:13 root [INFO]: Snapshot Id: c2794df8-4a48-44a1-8b45-03904ce6a9c7
03/05/2022 14:35:13 root [INFO]: Patch Baseline: {'accountId': '075727635805', 'baselineId': 'pb-0c7e89f
03/05/2022 14:35:13 root [INFO]: Reboot Option: NoReboot
03/05/2022 14:35:13 root [INFO]: Unable to initialize exit code reporting: No metrics ID from server
03/05/2022 14:35:13 root [WARNING]: Unable to locate yum, code: 1.

```

Step 7: install Patches On-Demand

18. From the **Systems Manager Page**:

- a. Select **Patch Manager** under **Node management**.
- b. Select **Patch now**.
- c. From the **Patch instances now page**:
 - i. **Patching operation**: Select **Scan and Install**.
 - ii. **Reboot option**: Select **Reboot if needed**.
 - iii. **Instances to patch**: Select **Patch only the target instances I specify**.
 - iv. **Target selection**: Select **Choose instances manually**.
 - a) **Instances**: Select **Windows Server 2019** server.
 - vi. **Patching log storage**: Dropdown and select **ec2ssm-123** S3 bucket.
 - vii. Select **Patch now**.

Note: You can see the **Execution details are loading**.

Note: **Wait**, till **Operation gets Successful**.

Step 8: View the Patches Install Status

19. From the **Systems Manager** Page:

- Select **Patch Manager** under **Node management**.
- Select **Dashboard**.

Note: You can see the **Patch operation history** details.

- Select the **Install** to view the **status**.

Patch operations history
This summary of recent patching operations indicates whether an operation was started manually, or started by a maintenance window or State Manager association. Choose an operation link to view the command output.

Filter tasks

Patch operation	Started by	Document name	End time	Status	Targets
Install	Association	AWS-RunPatchBaselineWithHooks	5 March 2022, 21:29 GMT+5:30	Success	InstanceIds: i-0675d4d964245896f
Scan	Association	AWS-RunPatchBaseline	5 March 2022, 20:15 GMT+5:30	Success	InstanceIds: i-0675d4d964245896f, i-0fe043c86ab8c0211

- Select the **Instance Id**.
- Select the **View Output**.

Command ID: 2b530ba0-f55a-46c4-8626-df9f3aa0bce9

Cancel command Rerun Copy to new

Command status

Overall status Success	Detailed status Success	# targets 1	# completed 1	# error 0	# delivery timed out 0
---------------------------	----------------------------	----------------	------------------	--------------	---------------------------

Targets and outputs

View output

Instance ID	Instance name	Status	Detailed Status	Start time	Finish time
i-0675d4d964245896f	EC2AMAZ-L2AT4JD.WORKGROUP	Success	Success	Sat, 05 Mar 2022 15:59:12 GMT	Sat, 05 Mar 2022 15:59:33 GMT

- Expand** the **Output** under **PreInstallScan** step name.

Step 1 - Command description and status

Status

Success

Detailed status

Success

Step name

PreInstallScan

Start time

Sat, 05 Mar 2022 15:59:12 GMT

▼ Output

The command output displays a maximum of 48,000 characters. You can view the complete command output in [Amazon S3](#).

```

OtherNonCompliantCount : 0
NotApplicableCount : 1901
UnreportedNotApplicableCount : 0

EC2AMAZ-L2AT4JD - PatchBaselineOperations Assessment Results - 2022-03-05T15:59:32.756

Scan found no missing updates.

```

b) **Expand** the **Output** under **VerifyLocalPatchStatesWindows** step name.

Step 3 - Command description and status

Status

Success

Detailed status

Success

Step name

VerifyLocalPatchStatesWindows

Start time

Sat, 05 Mar 2022 15:59:33 GMT

▼ Output

The command output displays a maximum of 48,000 characters. You can view the complete command output in [Amazon S3](#).

```

No missing patches were detected. Exiting document execution

Step exited with code 168. Therefore, marking step as succeeded. Further document steps will be skipped.

```

Task 5: Scan and Install the Patches On-Schedule

In this task, you will use AWS Systems Manager Maintenance Windows let you define a schedule for patching an operating system. Each Maintenance Window has a schedule, a duration, a set of registered targets, and a set of registered tasks. Typically, you want to apply your patches at a time when there is the least impact to your organization.

Step 1: Create Linux Instance

20. In the **AWS Management Console**, on the **Services** menu, click **EC2**.

21. Click **Instance**.

22. Click **Launch Instance**.

a. In the **Choose an Amazon Machine Image (AMI)** page:

- i. Select the **Amazon Linux 2 AMI [Kernel 5.x]**.
- ii. Click **Select**.

b. In the **Choose an Instance Type** page:

- i. Select the **t2.micro**.
- ii. Click **Next: Configure Instance Details**.

c. In the **Configure Instance Details** page:

- i. **IAM Role:** Dropdown and select **EC2-SSM-Role**.

Note: Leave the other settings as default.

- ii. Select **Next: Add Storage**.

d. In the **Add Storage** page:

Note: Leave all the settings as default.

- i. Select **Next: Add Tags**.

e. In the **Add Tags** page:

- i. Click on **Add Tag**:
 - a) **Key:** Write **Name**.

- b) **Value:** Write **Linux Server 2**.
 - ii. Click on **Add Tag**:
 - a) **Key:** Write **Project**.
 - b) **Value:** Write **HRMS**.
 - iii. Select **Next: Configure Security Group**.
- f. In the **Configure Security Group** page:
 - i. **Assign a security group:** Select **Create a new security group**.
 - a) **Security group name:** Write **Linux-Server-SG**.
 - b) **Description:** Write **Linux Server Security Group**.

Info: SSH Port [22], already added and allowed from anywhere [0.0.0.0/0].

Note: Leave the other settings as default.

- ii. Select **Review and Launch**.
- g. In the **Review Instance Launch** page:
 - i. Select **Launch**.
 - a) In the **Select an existing key pair or create a new key pair** section:
 - 1) **First selection box:** Dropdown and select **Choose an existing key pair**.
 - 2) **Key pair name:** Dropdown and select **My-SEC-LAB-KP**.
 - 3) **Enable** the **Checkmark** against the ***I acknowledge that I have access***

Note: Leave the other settings as default.

- ii. Click on **Launch Instances**.
- h. Select the **View Instances**.

Step 2: Check the Linux Server Status

- 23. In the **AWS Management Console**, on the **Services** menu, click **EC2**.
- 24. Click **Instance**.
- 25. Select **Linux Server 2**.
 - a. **Wait** for the **Instance State** to change to **Running** state.
 - b. **Wait** for the **Status check** to change to **2/2 checks passed**.

Note: **Wait** (~5 mnts.) for the server deployment.

Step 3: Create Maintenance Window

- 26. In the **AWS Management Console**, on the **Services** menu, click **Systems Manager**.
 - a. Select **Maintenance windows** under **Change management**.
 - i. Select **Create maintenance window**.
 - a) In the **Provide maintenance window details** section:
 - 1) **Name:** Write **HRMS-Production-Window**.



The screenshot shows the 'Provide maintenance window details' section of the AWS Management Console. It contains a form with a 'Name' field. The text 'Name' is followed by the instruction 'Type a name for this maintenance window.' The input field contains the text 'HRMS-Production-Window'. Below the input field, there is a validation message: 'It has to be between 3 and 128 characters. Valid characters contain the following: a-z, A-Z, 0-9, and _-'. The entire form is enclosed in a black border.

- b) In the **Schedule** section:

- 1) **Specify with:** Select **Cron schedule builder** and select **Daily**.
- 2) **Days:** Dropdown and select **Current Day**.
- 3) **Time:** Type **Current time** in **hh:mm** hours (**add 10 mnts.** from **current time**)

- 4) **Duration:** Write **1** hours.
- 5) **Stop initiating tasks:** Write **0** hour.

- 6) **Schedule time zone:** Dropdown and select your **Current time zone**.

- c) Select **Create maintenance window**.

Step 4: Scan and Install the Patches On-Schedule

27. From the **Systems Manager** Page:

- a. Select **Patch Manager** under **Node management**.
- b. Select **Configure patching**.
 - i. In the **Instances to patch** section:
 - a) **How do you want to select instances:** Select **Enter instance tags**.
 - b) **Instance tags:**
 - 1) **Tag key:** Write **Project**.
 - 2) **Tag value:** Write **HRMS**.
 - 3) Select **Add**.

Instances to patch

How do you want to select instances?

☒ Enter instance tags

☐ Select a patch group

☐ Select instances manually

Instance tags

Specify one or more instance tag key/value pairs to identify the instances you want to patch.

Project HRMS Add

Enter a tag key and optional value applied to the instances you want to target, and then choose Add.

Note: You can see the added **Tag** [Project:HRMS].

Instance tags

Specify one or more instance tag key/value pairs to identify the instances you want to patch.

Tag key Tag value (optional) Add

Enter a tag key and optional value applied to the instances you want to target, and then choose Add.

Project : HRMS X

- ii. In the **Patching schedule** section:
 - a) **How do you want to specify a patching schedule:**
Select **Schedule an existing Maintenance Window**.

- 1) **Maintenance Window:** Dropdown and select **HRMS-Production-Window**.

Patching schedule

How do you want to specify a patching schedule?

☒ Select an existing Maintenance Window
☐ Schedule in a new Maintenance Window
☐ Skip scheduling and patch instances now

Maintenance Window

Select a Maintenance Window [↗](#)

HRMS-Production-Window (ID: mw-054037b281d419bc9) ▼

- iii. In the **Patching operation** section:

- a) Select **Scan and install**.

Patching operation

☒ Scan and install
Scans each target instance and compares its installed patches with the list of approved patches in the patch baseline. Downloads and installs all approved patches that are missing from the instance.

☐ Scan only
Scans each target instance and generates a list of missing patches for you to review.

- c. Select **Configure patching**.

Step 5: View the Patch Operation Details

28. From the **Systems Manager** Page:

- a. Select **Patch Manager** under **Node management**.
- b. Select **Dashboard**.
- c. In the **Recurring patching tasks** section:

Note: You can see the **PatchingTasks** details.

Recurring patching tasks

The following is a list of State Manager associations and maintenance windows that run any patching-related task. Choose a task name to view its details

Patching task name	Task type	Document name	Schedule
AWS-PatchNowAssociation ↗	Association	AWS-RunPatchBaselineWithHooks	-
PatchingTask ↗	MW Task	AWS-RunPatchBaseline	cron(0 40 0 ? * SUN *)

d. In the **Patch operation history** section:

Note: You can see the **Patch operations history** details.

Note: Wait, till new **Install Patch operation initiated** as per **maintenance Window**.

Patch operations history					
This summary of recent patching operations indicates whether an operation was started manually, or started by a maintenance window or State Manager association. Choose an operation link to view the command output.					
Filter tasks					
Patch operation	Started by	Document name	End time	Status	Targets
Install	Other	AWS-RunPatchBaseline	6 March 2022, 01:05 GMT+5:30	InProgress	tag:Project: HRMS
Install	Association	AWS-RunPatchBaselineWithHooks	5 March 2022, 21:29 GMT+5:30	Success	InstanceIds: i-0675d4d964245896f

i. Click on the **Install** against **InProgress** status:

Note: You can see the **Command status** is **In Progress**.

Command status					
Overall status In Progress	Detailed status In Progress	# targets 2	# completed 0	# error 0	
Targets and outputs					
Instance ID	Instance name	Status	Detailed Status	Start time	
i-0fe043c86ab8c0211	ip-172-31-88-194.ec2.internal	In Progress	In Progress	Sat, 05 Mar 2022 19:35:50 GMT	
i-0b2f030515b78cfc1	ip-172-31-25-64.ec2.internal	Success	Success	Sat, 05 Mar 2022 19:35:50 GMT	

Note: Wait, till **Overall status** shows **Success**. **Keep Refresh** your screen unless you can see the **Overall status** shows **Success**.

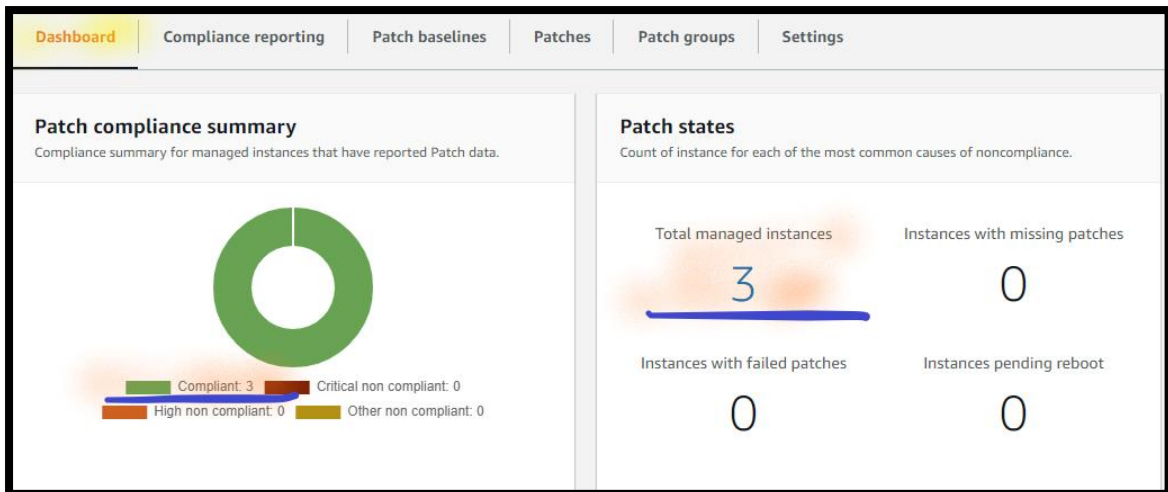
Command status					
Overall status Success	Detailed status Success	# targets 2	# completed 2		

Step 6: View the Patches Compliance

29. From the **Systems Manager** Page:

- Select **Patch Manager** under **Node management**.
- Select **Dashboard**.

Note: You can see the **Patch states** details.



- Select **Compliance Reporting**.

Note: You can see the **Compliance status** details.

Dashboard	Compliance reporting	Patch baselines	Patches	Patch groups	Settings
Instance patching summary (3)					
Search					
	Name	Instance ID	Compliance sta...	Baseline ID used	
☐	Windows ...	i-0675d4d964245896f 🔗	✓ Compliant	pb-09ca3fb51f0412ec3 🔗	
☐	Linux 2 Se...	i-0b2f030515b78cfc1 🔗	✓ Compliant	pb-0be8c61cde3be63f3 🔗	
☐	Ubuntu S...	i-0fe043c86ab8c0211 🔗	✓ Compliant	pb-0c7e89f711c3095f4 🔗	

Task 5: Customize Patch Baseline

In this task, you can create your own custom patch baselines. You can also choose which patches to auto-approve.

For each auto-approval rule that you create, you can specify an auto-approval delay. This delay is the number of days to wait after the patch was released, before the patch is automatically approved for patching.

Step 1: Create Patch Baseline

30. From the **Systems Manager Page**:

- a. Select **Patch Manager** under **Node management**.
- b. Select **Patch baselines**.
- c. Select **Create Patch baseline**.
 - i. In the **Patch baseline details** section:
 - a) **Name**: Write **Windows-Customise-Patch**.
 - b) **Description**: Write **Windows Customise Patch Baseline**.
 - c) **Operating System**: Dropdown and select **Windows**.

Name

Windows-Customise-Patch

You can use letters, numbers, periods, dashes, and underscores in the name.

Description - optional

Windows Customise Patch Baseline

Operating system

Select the operating system you want to specify approval rules and patch exceptions for.

Windows

Default patch baseline

☐ Set this patch baseline as the default patch baseline for Windows Server instances.

Note: Leave the other settings as default.

- ii. In the **Approval rules for operating systems** section:
- a) **Product:** Dropdown and select **WindowsServer 2019**.
 - b) **Severity:** Dropdown and select **All**.
 - c) **Classification:** Dropdown and select **All**.
 - d) Auto-approval: Select **Approve patches after a specified number of days**.
 - 1) **Specify the number of days:** Wrote **0** days.
 - e) **Compliance reporting:** Dropdown and select **Critical**.

The screenshot shows a configuration window for 'Approval rules for operating systems'. It has a 'Remove rule' button in the top right. The 'Product' section has a dropdown 'Select patches by product' with 'All' selected. The 'Severity' section has a dropdown 'Select patches by severity' with 'All' selected. The 'Classification' section has a dropdown 'Select patches by classification' with 'All' selected. The 'Auto-approval' section has a radio button selected for 'Approve patches after a specified number of days', with a text input '0 days' below it. The 'Compliance reporting - optional' section has a dropdown 'Specify the severity level to report for patches that match this rule.' with 'Critical' selected.

- iii. In the **Approval rules for applications** section:

Note: Leave the all settings as default.

- iv. In the **Patch exceptions** section:

Note: Leave the all settings as default.

- v. Select **Create patch baseline**.

Step 2: Configure Patch group

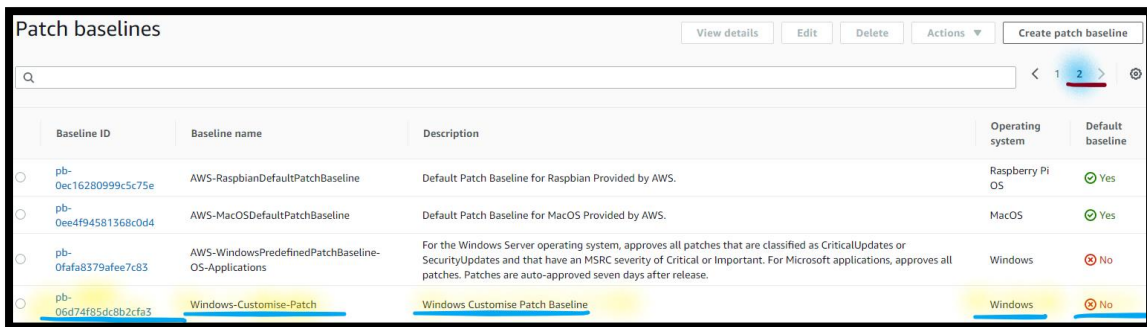
31. From the **Systems Manager** Page:

- a. Select **Patch Manager** under **Node management**.
- b. Select **Patch baselines**.

Note: You can see your **Customise Patch Baseline**.

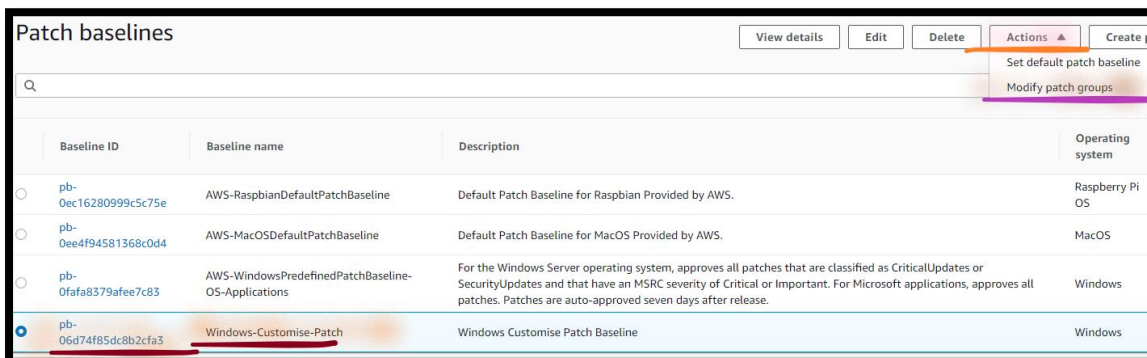
Note: You can **Go to next page** in Patch baseline, if you can't see your **Customise Patch Baseline**.

- i. **Copy** the **Baseline ID** Of the **Windows-Customise-Patch** in the **Notepad**.



Baseline ID	Baseline name	Description	Operating system	Default baseline
pb-0ec16280999c5c75e	AWS-RaspbianDefaultPatchBaseline	Default Patch Baseline for Raspbian Provided by AWS.	Raspberry Pi OS	Yes
pb-0ee4f94581368c0d4	AWS-MacOSDefaultPatchBaseline	Default Patch Baseline for MacOS Provided by AWS.	MacOS	Yes
pb-0fafa8379afee7c83	AWS-WindowsPredefinedPatchBaseline-OS-Applications	For the Windows Server operating system, approves all patches that are classified as CriticalUpdates or SecurityUpdates and that have an MSRC severity of Critical or Important. For Microsoft applications, approves all patches. Patches are auto-approved seven days after release.	Windows	No
pb-06d74f85dc8b2cfa3	Windows-Customise-Patch	Windows Customise Patch Baseline	Windows	No

- c. Select **Windows-Customise-Patch**.
 - i. Select **Actions**.
 - ii. Select **Modify patch groups**.



Baseline ID	Baseline name	Description	Operating system
pb-0ec16280999c5c75e	AWS-RaspbianDefaultPatchBaseline	Default Patch Baseline for Raspbian Provided by AWS.	Raspberry Pi OS
pb-0ee4f94581368c0d4	AWS-MacOSDefaultPatchBaseline	Default Patch Baseline for MacOS Provided by AWS.	MacOS
pb-0fafa8379afee7c83	AWS-WindowsPredefinedPatchBaseline-OS-Applications	For the Windows Server operating system, approves all patches that are classified as CriticalUpdates or SecurityUpdates and that have an MSRC severity of Critical or Important. For Microsoft applications, approves all patches. Patches are auto-approved seven days after release.	Windows
pb-06d74f85dc8b2cfa3	Windows-Customise-Patch	Windows Customise Patch Baseline	Windows

- a) **Patch groups:** Write **ERP-Prod**.

Patch groups

You can create up to 25 tag values to define patch groups for this patch baseline. Tag keys are automatically named **Patch Group**. [Learn more](#)

Baseline ID
pb-06d74f85dc8b2cfa3

Baseline name
Windows-Customise-Patch

Baseline description
Windows Customise Patch Baseline

Patch groups

ERP-Prod

Patch group values can consist of up to 256 letters, numbers, and the following characters: . _ + @ / - + :

No patch groups attached

b) Select **Add**.

Note: You can see **ERP_Prod** added in the Patch groups.

Baseline name
Windows-Customise-Patch

Baseline description
Windows Customise Patch Baseline

Patch groups

Patch group values can consist of up to 256 letters, numbers, and the following characters: . _ + @ / - + :

ERP-Prod X

c) Select **Close**.

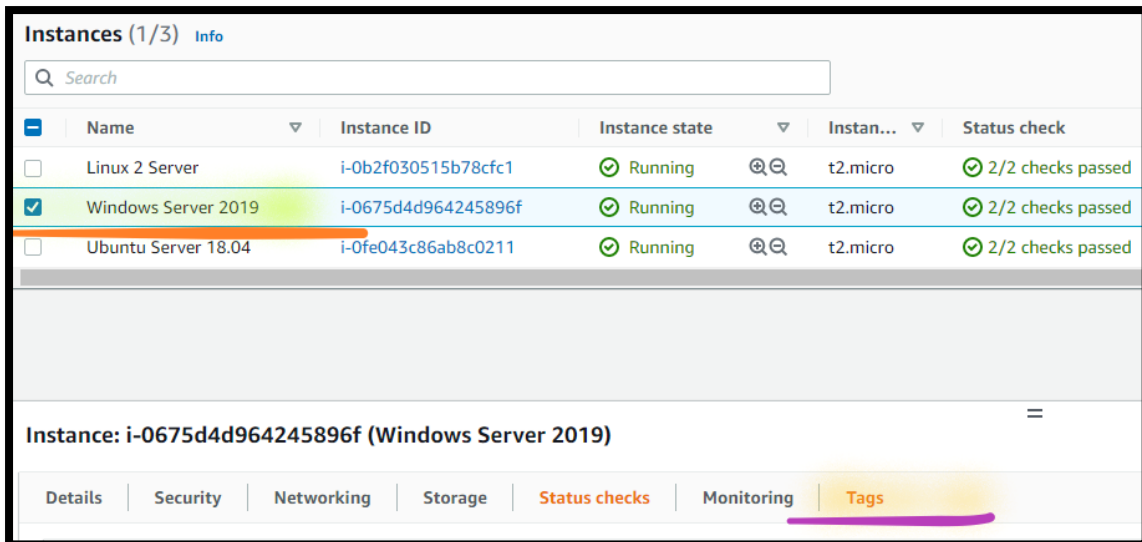
Step 3: Associate Instance with Patch group

32. In the **AWS Management Console**, on the **Services** menu, click **EC2**.

33. Click **Instance**.

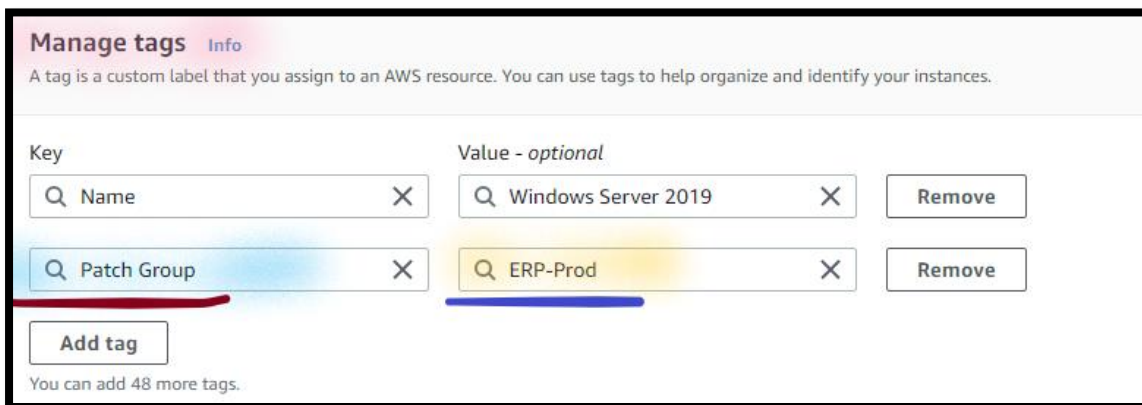
34. Select **Windows Server 2019**.

a. **Go below in the console**, select **Tags**.



b. From the **Tags** section:

- i. Select **Manage tags**.
- ii. Click on **Add tag**:
 - a. **Key**: Write **Patch Group**.
 - b. **Value**: Write **ERP-Prod**.



c. Select **Save**.

Step 4: Scan and Install the Patches using Patch group

35. From the **Systems Manager** Page:

- a. Select **Patch Manager** under **Node management**.
- b. Select **Configure patching**.
 - i. In the **Instances to patch** section:

- a) **How do you want to select instances:** Click on **Select a patch group**.
- b) **Patch groups:** Dropdown and select **ERP-Prod**.
- ii. In the **Patching schedule** section:
 - a) **How do you want to specify a patching schedule:**
Select **Skip scheduling and patch instances now**.

The screenshot shows a web interface with two main sections. The first section, titled 'Instances to patch', contains a question 'How do you want to select instances?' with three radio button options: 'Enter instance tags', 'Select a patch group' (which is selected and highlighted with a green circle), and 'Select instances manually'. Below this is a 'Patch groups' section with the instruction 'Specify one or more patch groups to identify the instances you want to patch.' It features a dropdown menu labeled 'Select patch group' and a tag labeled 'ERP-Prod' with a close button (X). The second section, titled 'Patching schedule', contains a question 'How do you want to specify a patching schedule?' with three radio button options: 'Select an existing Maintenance Window', 'Schedule in a new Maintenance Window', and 'Skip scheduling and patch instances now' (which is selected and highlighted with a blue circle).

- iii. In the **Patching operation** section:
 - a) Select **Scan and install**.
- c. Select **Configure patching**.

Step 5: View the Patch Operation Details

36. From the **Systems Manager Page**:

- a. Select **Patch Manager** under **Node management**.
- b. Select **Dashboard**.
- c. In the **Patch operation history** section:

Note: You can see the **Patch operations history** details.

Note: You can see **Install Patch operation initiated** for Patch group.

Patch operations history
This summary of recent patching operations indicates whether an operation was started manually, or started by a maintenance window or State Manager association. Choose an operation link to view the command output.

Filter tasks

Patch operation	Started by	Document name	End time	Status	Targets
Install	Other	AWS-RunPatchBaseline	6 March 2022, 16:06 GMT+5:30	InProgress	tag:Patch Group: ERP-Prod
Install	Association	AWS-RunPatchBaselineWithHooks	5 March 2022, 21:29 GMT+5:30	Success	InstanceIds: i-0675d4d964245896f

i. Click on the **Install** against **InProgress** status:

Note: You can see the **Command status** is **In Progress**.

Command status

Overall status In Progress	Detailed status In Progress	# targets 1	# completed 0	# error 0
-------------------------------	--------------------------------	----------------	------------------	--------------

Note: Wait, till **Overall status** shows **Success**. **Keep Refresh** your screen unless you can see the **Overall status** shows **Success**.

Command status

Overall status Success	Detailed status Success	# targets 1	# completed 1
---------------------------	----------------------------	----------------	------------------

Step 6: View the Patches Compliance

37. From the **Systems Manager** Page:

- Select **Patch Manager** under **Node management**.
- Select **Dashboard**.

Note: You can see the **Patch states** details.

- c. Select **Compliance Reporting**.

Note: You can see the **Compliance status** details.

- i. Select **Windows Server 2019**.

Note: Verify **Patch Baseline ID** used for Patching.

- ii. Select **View logs**.

Instance patching summary (3) View log

Search

	Name	Instance ID	Compliance sta...	Baseline ID used	Patch group used
☒	Windows Server 2019	i-0675d4d964245896f	Compliant	pb-06d74f85dc8b2cfa3	ERP-Prod
☐	Linux 2 Server	i-0b2f030515b78cfc1	Compliant	pb-0be8c61cde3be63f3	
☐	Ubuntu Server 18.04	i-0fe043c86ab8c0211	Compliant	pb-0c7e89f711c3095f4	

- a) **Expand** the **Output** under **PatchWindows** step name.

Status Success Detailed status Success

Step name PatchWindows Start time Sun, 06 Mar 2022 10:36:21 GMT

Output

The command output displays a maximum of 48,000 characters. You can view the complete command output in either Amazon S3 or CloudWatch Logs, if you have access.

```

InstalledCount : 33
InstalledRejectedCount : 0
InstalledPendingRebootCount : 0
InstalledOtherCount : 6
FailedCount : 0
MissingCount : 0
  
```

Task 6: Configure AWS Inspector

In this task, you can set up Amazon Inspector to run security assessment scans on a target resource group.

Step 1: Enable Inspector

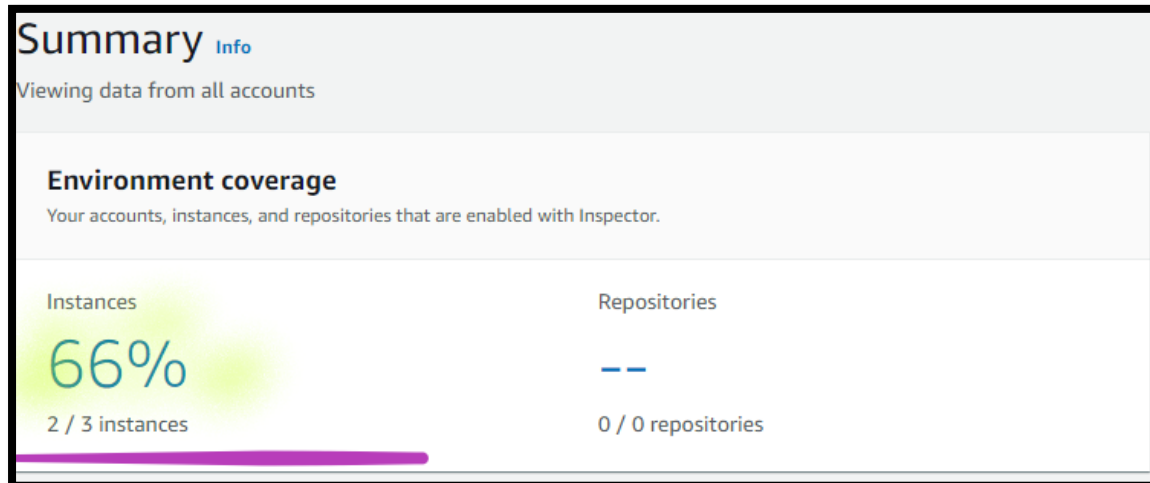
38. Dropdown and select **US East (N. Virginia)** region.

39. In the **AWS Management Console**, on the **Services** menu, click **Inspector**.

- Select **Get started**.
- Select **Enable Inspector**.
- Go to left**, select **Dashboard**.

Note: If you are not seeing Dashboard option, click on **three lines** on the left side.

Note: You can see the **Instances covered** by **AWS Inspector**.



Info: Amazon Inspector uses the AWS Systems Manager (SSM) agent for scanning.

Step 2: Verify Covered Instances details

40. From the **AWS Inspector** console:

- Select **Account management** under **settings**.
- Select **Instances**.

Note: You can see the Instances covered by AWS Inspector.

Note: You can see the **Windows OS** as **Unspported OS**.

Accounts

Instances

Repositories

Images

All

3

Scanning

2

Not scanning

1

Instances (3)

Info

▼

●

Resource type

EQUALS

AWS EC2 Instance

⊗

Add filter

EC2 instance

EC2 instance name

Account

AMI

Operating system

Status

i-0675d4d964245896f

Windows Server 2019

Cloud(504340965905)

ami-0c19f80dba70861db

WINDOWS

Unspported OS

i-0b2f030515b78cfc1

Linux 2 Server

Cloud(504340965905)

ami-0c293f3f676ec4f90

LINUX

Scanning

i-0fe043c86ab8c0211

Ubuntu Server 18.04

Cloud(504340965905)

ami-0e472ba40eb589f49

LINUX

Scanning

Step 3: View the Findings

41. From the **AWS Inspector** console:

- Select **By vulnerability** under **Findings**.

Note: You can see the Vulnerabilities details.

By vulnerability

By instance

By container image

By repository

All findings

By vulnerability (8)

Choose a row to view the vulnerability's details and associated findings.

Add filter

Vulnerability	Critical	High	All
Port 3389 is reachable from an Internet Gateway	0	0	1
Port 22 is reachable from an Internet Gateway	0	0	2
CVE-2021-44731 - snapd	0	1	1
CVE-2021-44730 - snapd	0	1	1
CVE-2021-4120 - snapd	0	0	1
CVE-2021-3155 - snapd	0	0	1
CVE-2019-7306 - byobu	0	0	1
CVE-2017-9525 - cron	0	0	1

i. Open **Port 3389 Vulnerability**.

Note: You can see the **vulnerability details** and **impacted resources**.



Port 3389 is reachable from an Internet Gateway

Vulnerability

Details Info

Solution summary

You can restrict access to your instance by modifying the Security Groups or ACLs in the network path.

Findings (1) Info

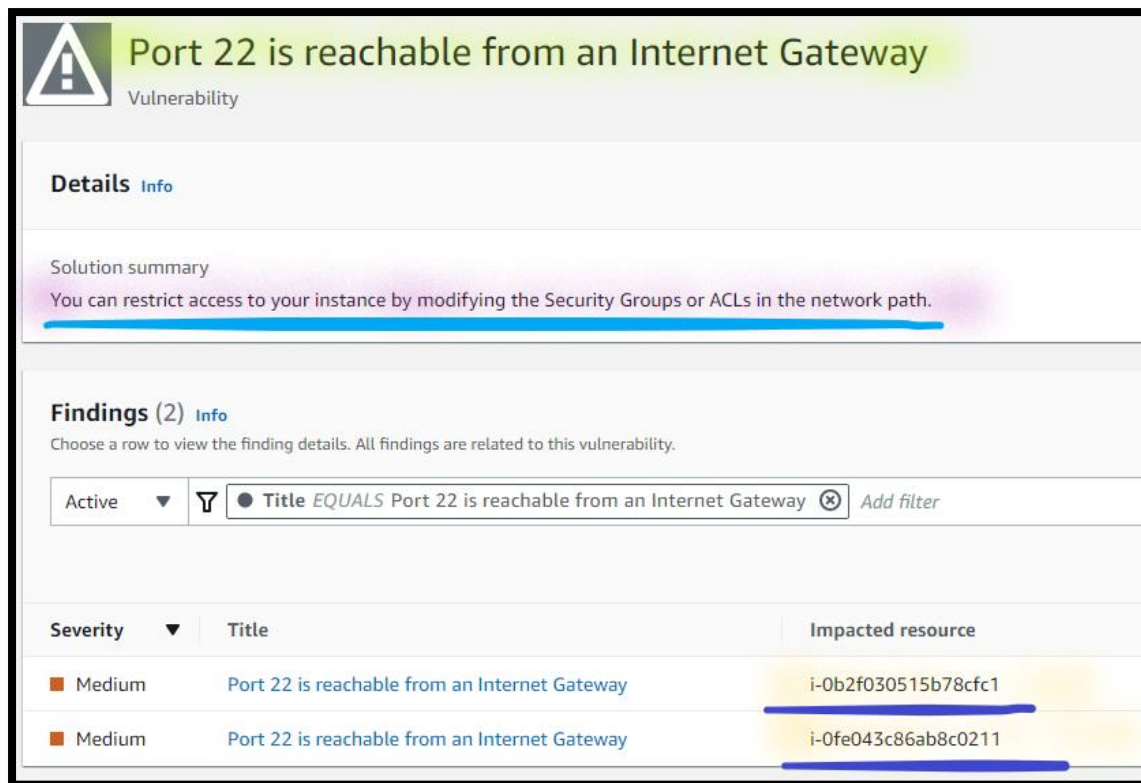
Choose a row to view the finding details. All findings are related to this vulnerability.

Active Add filter

Severity	Title	Impacted resource
Medium	Port 3389 is reachable from an Internet Gateway	i-0675d4d964245896f

ii. Open **Port 22 Vulnerability**.

Note: You can see the **vulnerability details** and **impacted resources**.



Port 22 is reachable from an Internet Gateway
Vulnerability

Details Info

Solution summary
You can restrict access to your instance by modifying the Security Groups or ACLs in the network path.

Findings (2) Info
Choose a row to view the finding details. All findings are related to this vulnerability.

Active ▼ Add filter

Severity	Title	Impacted resource
Medium	Port 22 is reachable from an Internet Gateway	i-0b2f030515b78cfc1
Medium	Port 22 is reachable from an Internet Gateway	i-0fe043c86ab8c0211

- b. Select **All findings** under **Findings**.

Note: You can see the **vulnerabilities details** with **Severity**.

Task 7: Cleanup the Environment

Step 1: Delete Customize Patch Baseline

42. From the **Systems Manager** Page:

- Select **Patch Manager** under **Node management**.
- Select **Patch baselines**.
- Select **Windows-Customise-Patch**.
 - Select **Actions**.
 - Select **Modify patch groups**.
 - Select **Remove** **ERP-Prod** from **Patch groups**.
 - Select **Close**.

- d. Select **Windows-Customise-Patch**.
 - i. Select **Delete**.
 - ii. Select **Delete**.

Step 2: Delete the EC2SSM-123 bucket

- 43. In the **AWS Management Console**, on the **Services** menu, click **S3**.
- 44. Select **Buckets**.
- 45. Select **ec2ssm-123** bucket.
 - a. Select **Empty**.
 - b. Type **permanently delete** to **Delete all the objects**.
 - c. Select **Empty**.
 - d. Select **Exit**.
- 46. Select **ec2ssm-123** bucket.
 - a. Select **Delete**.
 - b. Type **ec2ssm-123** bucket name, to **Delete bucket**.
 - c. Select **Delete bucket**.

Step 3: Terminate EC2 Instances

- 47. In the **AWS Management Console**, on the **Services** menu, click **EC2**.
- 48. Click **Instances**.
- 49. Select **All the Three** instances.
 - a. Click on **Instance state**.
 - b. Select **Terminate instance**.
 - c. Select **Terminate**.

Step 4: Enable Inspector

50. In the **AWS Management Console**, on the **Services** menu, click **Inspector**.

- a. Select **General** under *settings*.
- b. Select **Disable Inspector**.
 - i. Type **Disablee** to *disable inspector*.
 - ii. Select **Disable Inspector**.